PRAVAAH

Ward-level water scarcity prediction for Bengaluru

Tech rundown & pitch prep for the team

What we built, the tech stack, how every feature works, the numbers to quote, a demo script, and answers to the questions judges are likely to ask.

SDG 6: Clean Water and Sanitation
Repository: github.com/jyothsnasivakumar/pravaah (private)

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1. Pravaah in one minute

Bengaluru ran out of water in early 2024: thousands of borewells dried up and tanker prices spiked, hitting the outer wards (Whitefield, Mahadevapura, RR Nagar, Bommanahalli) hardest. Existing dashboards show what already happened, for the whole city. Nobody tells a resident of Whitefield, weeks ahead, that **their** area is about to run dry, or why.

Pravaah predicts, for each of Bengaluru's **243 BBMP wards**, in how many days a water shortage is expected (as a range, e.g. "37–42 days"), explains **why** in plain language, and shows what residents and officials can do about it, in **English and Kannada**.

Our six USPs

1. **Ward-level predictions**, not one vague number for "Bengaluru Urban".
2. **Explainable**: every prediction comes with its causes and how much each one matters. No black box.
3. **Interactive risk map** with a scarcity heatmap and index maps.
4. **Bilingual chatbot (JalaMitra)** that answers "why" and "what can I do" in English or Kannada.
5. **Backtested on the 2024 crisis**: rewound to 1 Dec 2023, the model flagged Whitefield 91 days before the crisis peaked.
6. **Two views**: a simple citizen view and an official view with rankings and an intervention planner.

2. Numbers to quote in the pitch

All figures come from the running app. Use them exactly as written.

What	Number
Coverage	243 BBMP wards (2023 delimitation), every ward forecast separately
Rainfall history	Daily rainfall, Jan 2014 – Sep 2026, 16 points over the city + 2 in the Cauvery catchment
Why 2024 happened	2023 rainfall over Bengaluru: 732 mm (driest year in our record). Kabini catchment: 1,186 mm vs ~2,000 mm normal
Headline prediction	On 1 Jan 2024 the model said **Whitefield: shortage in 37–42 days** (≈ 7–12 Feb 2024). Mahadevapura: 46–52 days
Early warning	On 1 Dec 2023 (data available that day only) Whitefield was flagged high-risk: shortage in 68–98 days, i.e. **91 days before the early-March 2024 crisis peak**
Model behaves sensibly	Wards in shortage: ~2–4% in May of normal years (2022, 2023) vs **30% on 1 Mar 2024 and 37% by mid-April 2024**
Why Whitefield	Main causes: ~67% of water from borewells and rain 28% below normal. IT/commercial load brings shortage 76 days sooner
Intervention planner	+30% Cauvery pipe coverage and +50% rainwater recharge moves Whitefield from "37–42 days" to **low risk**

What	Number
Today (30 Sep 2026)	52 of 243 wards at high or critical risk within 90 days, ~27 lakh people (estimated)
Forecast ensemble	~30 scenarios per forecast (past rainfall years × 3 demand levels)

Before the pitch: the "crisis peak, early March 2024" reference date is based on news coverage from memory. Find and cite one news article (e.g. on borewells drying up in March 2024) so we can show a source if asked.

3. Tech stack

Layer	Technology	Why we chose it
Rainfall data	ERA5 reanalysis via the Open-Meteo archive API	Free, no API key, global, daily since 1940. Lets us scale beyond Bengaluru without new data deals
Ward boundaries	BBMP ward GeoJSON (datameet / KGIS)	Official 243-ward map, open data
Model	Python 3 + NumPy	Fast vectorised simulation: all wards × 30 scenarios × 365 days in under half a second
API / backend	FastAPI (Python)	Simple, fast, auto-documented REST API; serves the built website too
Chatbot	Claude Haiku 4.5 (Anthropic API) + built-in bilingual answer engine	Good Kannada, fast and cheap. Works without a key using the built-in engine
Frontend	React 19 + Vite	Component-based UI, very fast builds
Maps	Leaflet (react-leaflet), Esri street tiles	Free open-source mapping; choropleth and street-map layers
Charts	Recharts	Groundwater forecast bands and backtest charts
Code & hosting	GitHub + Vercel	Website as static files; model runs as a Python serverless function. Auto-deploys on every push

How the pieces connect

1. **Data in:** backend/fetch_data.py downloads rainfall into backend/data/rainfall.json; ward shapes live in backend/data/bbmp_wards.geojson.
2. **Model:** backend/model.py simulates groundwater for every ward, every day, and produces forecasts and explanations.
3. **API:** backend/main.py exposes it at /api/... (predictions, ward details, backtest, interventions, chat).
4. **Chatbot:** backend/chat.py (Claude or built-in answers); backend/content.py holds the English/Kannada advice text.
5. **Website:** frontend/src/ (React). It calls the API and draws the map, panels, charts and tour.
6. **Deployment:** vercel.json + api/index.py tell Vercel how to build the site and run the Python API.

4. How the prediction works (in plain words)

Think of each ward as having an **underground water tank** (its aquifer). Every day, water goes in and out:

```
groundwater += rain recharge − borewell pumping + pipe leakage + inflow from the surrounding aquifer
```

Term	What it means and how we compute it
Rain recharge	A share of each day's rain (above 2 mm) soaks in. Paved, dense wards absorb less; wards near lakes absorb more
Demand	People × 135 litres/day (CPHEEO norm) + extra load for IT parks/industry in specific wards
Piped supply	Cauvery water covers part of demand: high in the old core, low in the 110 villages added in 2007. Supply drops when the Cauvery catchment gets poor rain. Stage V (Oct 2024) gradually raises coverage in outer wards
Borewell pumping	Whatever the pipes don't supply is pumped from the ground. As the aquifer empties, borewell yield falls
Pipe leakage	~30% of piped water leaks and seeps back into the ground (a real Bengaluru effect)
Regional inflow	Neighbouring aquifers slowly refill a ward, but less after a dry year, because the regional water table drops too
Shortage	Declared when the ward's groundwater falls below **30% of usable storage**: borewells start failing and tanker dependence spikes

From simulation to "37–42 days"

- The model runs every day from 2014 to "today" to get each ward's current groundwater level.
- It then replays the next 12 months about **30 times**: once for each past rainfall year, at three demand levels.
- Each run gives "days until shortage". The range shown is the **middle half** of those runs (25th–75th percentile). The colour band (risk level) comes from the middle value.

Risk level	Days until shortage	Colour
Critical	0–30	Red
High	31–90	Orange
Medium	91–180	Yellow
Low	more than 180	Green

Calibration

Model settings (aquifer size, inflow rate, leakage share, how the regional water table reacts to dry years) were tuned so that normal years stay calm (~2–5% of wards short in summer) and Feb–Apr 2024 shows a city-wide crisis (~30–37%). It also flags stress in 2017 after the 2016 drought, which matches history.

5. Feature by feature

5.1 Ward-level prediction

What it does

- Gives every ward its own "shortage in X–Y days" with approximate dates and a risk level.

How it works

- The simulation in section 4 runs for all 243 wards at once (NumPy arrays), so a full city forecast takes under half a second.
- Each ward has its own rainfall (nearest of 16 grid points), area, population estimate, pipe coverage and commercial load.

Where in the code

backend/model.py: predict(), _forecast()

> **Demo tip:** Search "Whitefield" in My Area with the date set to 1 Jan 2024: "37–42 days, around 7–12 Feb 2024".

5.2 Explainability: "Why?"

What it does

- Lists the causes of a ward's shortage, ranked, e.g. "Heavy dependence on borewells: MAIN CAUSE", "IT parks: 76 days sooner".

How it works

- **Counterfactual analysis**: we switch off one factor at a time (set it to the city-typical value), re-run the forecast, and measure how many days later the shortage arrives.
- Factors: rainfall deficit, borewell dependence, population density, commercial/IT load, Cauvery catchment deficit, paved surfaces.
- "Main cause" = without this factor there would be no shortage within a year.
- This is the same idea as SHAP values in machine learning, but computed directly on a physical model, so every number has a real-world meaning.

Where in the code

backend/model.py: FACTORS and the counterfactual loop in predict()

> **Demo tip:** Say: "We don't just predict, we can tell a resident exactly why, and the official exactly which lever to pull."

5.3 Interactive risk map and heatmap

What it does

- **Risk zones:** wards coloured by risk level on a real street map.
- **Heatmap:** a clean "Water Scarcity Index" map (0 = safe, 100 = shortage now) with a colour bar.
- **Groundwater** and **Borewell dependence** index maps in the same style.
- Hover any ward for its numbers; click to open it.
- **Timeline** under the map: drag or press ▶ to replay the forecast week by week from 2015 to today.

How it works

- The browser fetches ward shapes once and the forecast for the chosen date, then colours each ward with Leaflet.
- Each timeline position re-runs the model "as of" that date.

Where in the code

frontend/src/MapView.jsx, Timeline.jsx

> **Demo tip:** Switch to Heatmap at 1 Mar 2024 and press ▶ from late 2023: the red spreads across the outer wards.

5.4 JalaMitra chatbot (English + Kannada)

What it does

- Answers questions like "Why is there a shortage here?", "What can I do?", "When will it start?", "How do I harvest rainwater?", "How does this work?"

How it works

- Every answer is built from **that ward's actual prediction** (days, dates, rainfall vs normal, borewell share, causes).
- With an Anthropic API key it uses **Claude Haiku 4.5**, given the ward's data and instructions to answer in the chosen language.
- Without a key, a built-in engine detects the question type (why / when / what to do / rainwater / tankers / groundwater / accuracy) and writes a structured answer in English or Kannada.
- The language button in the header switches the whole site and the chatbot.

Where in the code

backend/chat.py, backend/content.py, frontend/src/Chat.jsx

> **Demo tip:** In the guided tour's last step it asks a Kannada question automatically.

5.5 Time Machine: backtesting the 2024 crisis

What it does

- Rewinds to any past date and shows what the model would have predicted then.
- Story card: "91 days of early warning" for Whitefield, with a dated timeline (flagged → groundwater crosses line → crisis peak).
- Chart of how many wards were at high/critical risk each week, Sep 2023 – Apr 2024.

How it works

- **No peeking at the future:** a backtest only uses rainfall that had already fallen, and its future scenarios only use earlier years.
- The 2024 backtest is precomputed so it loads instantly online.

Where in the code

backend/main.py: api_backtest(); frontend/src/BacktestView.jsx

> **Demo tip:** This is our strongest moment. Pause on "91 days of early warning".

5.6 Two views: Citizen and Official

What it does

- **My Area (citizen):** search your area → big countdown, groundwater gauge, reasons in plain words, action cards, "Ask JalaMitra".
- **Officials:** risk tiles, ward ranking (by soonest shortage, groundwater, borewell dependence, population), deep dive with groundwater history + forecast chart, model inputs labelled real/estimated.
- **Intervention planner:** sliders for Cauvery pipe extension, rainwater harvesting, demand cuts and tanker supply. The model re-runs all scenarios and shows before vs after.

How it works

- The planner sends the chosen levers to the API, which re-runs the forecast with those changes for that ward.

Where in the code

frontend/src/CitizenView.jsx, OfficialView.jsx; backend/main.py: api_intervene()

> **Demo tip:** Whitefield on 1 Jan 2024: +30% pipes and +50% recharge turns "37–42 days, high" into low risk.

5.7 Guided tour and Overview

What it does

- "▶ Take the tour" walks judges through all seven features in about a minute, switching view, ward, date, layer and language automatically and highlighting what to look at.
- Overview page: "X of 243 wards will run short within 90 days, affecting ~Y lakh people", most urgent wards, and a 3-step "How it works".

How it works

- Each tour step sets the app state, then spotlights one element.

Where in the code

frontend/src/Tour.jsx, Overview.jsx, App.jsx

> **Demo tip:** Open the site with ?tour=0 at the end of the URL to start the tour immediately.

6. What is real, what is estimated

Be upfront about this. Judges respect honesty more than fake precision, and the app labels it too.

Input	Status	Source / method
Ward boundaries (243)	Real	BBMP / KGIS via datameet
Daily rainfall 2014–today	Real	ERA5 reanalysis (Open-Meteo), 16 points over Bengaluru
Cauvery catchment rainfall	Real	ERA5 at Kodagu (KRS) and Wayanad (Kabini)
Stage V Cauvery commissioning	Real event	Oct 2024, public announcements
Ward population	Estimated	2011 census ÷ 243 (delimitation basis) × faster growth for outer wards
Cauvery pipe coverage	Estimated	High in the core, low in the 110 villages merged in 2007

Input	Status	Source / method
IT / industrial load	Estimated	Assigned to known hubs: Whitefield, Peenya, Bellandur, Mahadevapura…
Groundwater level	Simulated	Computed by the model; no well readings used yet

> Suggested line: "Our rainfall and ward data are real. Demand and groundwater are modelled estimates, clearly labelled. The next step is calibrating each ward against CGWB well measurements."

7. Suggested 3-minute demo

Open the site (Vercel link or http://localhost:8000) a minute early so it is warmed up, then click ▶ **Take the tour** and narrate:

Step	On screen	What to say
1	Risk map of all wards	"Every one of Bengaluru's 243 wards gets its own forecast."
2	Heatmap	"Here is where scarcity is closest. Red means shortage within weeks."
3	Whitefield, 1 Jan 2024	"A Whitefield resident would have seen: high risk, shortage in 37–42 days."
4	Why? list	"And why: 67% borewell dependence and rain 28% below normal. IT parks bring it 76 days sooner. No black box."
5	Time Machine	"Rewound to 1 Dec 2023, using only data from that day, we flagged Whitefield 91 days before the crisis peaked."
6	Planner	"Officials can test fixes before spending money: extending pipes plus recharge wells takes Whitefield to low risk."
7	Kannada chat	"And residents can ask in Kannada, and get answers grounded in their own ward's numbers."

Close with the impact line: early warning turns a crisis into planning, giving citizens time to store water and harvest rain, and officials time to schedule tankers and extend pipelines.

8. Questions judges may ask

Q: Is this real data or a mock-up?

Rainfall and ward boundaries are real downloaded data, 12 years of daily rainfall. Demand and pipe coverage are estimates and groundwater is simulated; the app labels each one. Swapping in official well and supply data is our next step.

Q: Is this AI / machine learning?

The core is a physics-style water balance model, chosen deliberately because it is explainable: every number maps to a real quantity. The explanations use counterfactual analysis (the idea behind SHAP). The chatbot uses a large language model (Claude). A machine-learning layer trained on real well data can sit on top once we have that data.

Q: How accurate is it?
We backtested on the 2024 crisis using only data available at the time: Whitefield was flagged on 1 Dec 2023, 91 days before the peak, and the model kept normal years calm. We are honest that settings were tuned on history including 2024, so the next step is validating on years and wells it has not seen.

Q: Why a range ("37–42 days") and not one number?
Future rain is uncertain. We replay ~30 scenarios from past years and show the middle half. A range is more honest and more useful for planning.

Q: Who would use this?
Residents and apartment associations (store water, book tankers early, harvest rain); BWSSB and BBMP (prioritise wards, schedule tankers, target pipeline and recharge investments).

Q: How does it scale beyond Bengaluru?
ERA5 rainfall is global and free; ward or village boundaries exist for every Indian city; CGWB monitors wells nationwide. The model only needs a boundary file and local demand estimates to run for a new city.

Q: What does it cost to run?
Almost nothing: open data, open-source software, and serverless hosting on Vercel. The AI chatbot costs fractions of a rupee per answer with Claude Haiku, and works without it.

Q: How is this different from existing dashboards (CGWB, KSNDMC)?
Those report past measurements, mostly city- or district-wide. Pravaah is forward-looking, ward-level, explains causes, lets officials test interventions, and talks to citizens in Kannada.

9. Known limitations and roadmap

Limitations (know these before the judges do)

- Groundwater is simulated, not measured; demand and pipe coverage are estimates.
- Settings were tuned with knowledge of the 2024 crisis, so the backtest is not a fully independent test.
- Some outer wards are flagged as chronically in shortage year after year; that may overstate risk there.
- Most wards come out either "safe for a year" or "short within weeks"; few sit in between.
- The chatbot uses built-in answers unless an Anthropic API key is set.

Roadmap for the 24-hour build and beyond

1. Calibrate each ward against **CGWB / Karnataka Groundwater Directorate well levels** (India-WRIS).
2. Add **BWSSB supply and consumption** data, **KSNDMC hobli rainfall** and **KRS/Kabini reservoir storage**.
3. Ward population from BBMP delimitation documents instead of an even split.

4. A machine-learning layer (gradient boosting + SHAP) trained on measured well data.
5. SMS / WhatsApp alerts in Kannada when a ward turns high-risk.
6. Expand to Mysuru, Hubballi–Dharwad and other Karnataka cities, then other states.

10. Running and sharing the app

Locally

```
cd backend
python -m venv .venv  &&  .venv\Scripts\activate
pip install -r requirements.txt
cd ../frontend  &&  npm install  &&  npm run build
cd ../backend  &&  uvicorn main:app --port 8000
```

Then open http://localhost:8000.

Online

Deployed on Vercel from the GitHub repo; every push to main redeploys automatically. Optional: add ANTHROPIC_API_KEY in Vercel → Settings → Environment Variables, then redeploy, to switch the chatbot to Claude.

Handy links (add to the end of the site address)

Add this	Opens
?tour=0	The guided tour of all features
?layer=heat&as_of=2024-03-01	Scarcity heatmap during the 2024 crisis
?view=area&ward=Whitefield&as_of=2024-01-01	Citizen view for Whitefield (add &lang=kn for Kannada)
?view=official&ward=Whitefield&as_of=2024-01-01	Official view
?view=backtest&ward=Whitefield&as_of=2023-12-01	Time Machine story